Transistor

Invented in 1947 at Bell Labs by John Bardeen, Walter Brattain and William Shockley, the transistor is a semiconductor device used to amplify and switch electronic signals and electrical powers. This little invention revolutionized the field of electronics and paved the way for cool things that we'd probably be unable to live without in 2013 such as the calculator, computer and radio. Why? Basically, a transistor is like a miniature on-off switch that allows a computer to process information. Computers can't operate without an integrated circuit (chip) and a chip can't operate without a transistor.

In the 1950s, the first transistor radio used only four transistors while in 1971, the first Intel chip used 2300. The latest chip has 820 million of these little miracles. Even more interesting is the fact that the original transistor could be held in your hand while current transistors are so small that more than two million 22nm tri-gate transistors could fit on the fullstop at the end of this sentence. Today, more than a billion Nobel prize winning transistors are manufactured across the globe every second.

Turbines

The turbine is almost single-handedly responsible for the technological advancement of the human race. Everything from the humble Windmill to the super-efficient turbo-jets in advanced fighter aircraft to even the super-massive cores of power stations is based on a turbine in some way or the other. The very nature of our existence is based on the amount of power we harness and the turbine is the only device that can harness it at the rate we require.

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The origins of this design are long lost (who actually did create the first windmill or water wheel?) but it was only with the invention of the